

# Materna: A Smart IoT-Based Monitoring and Alert System for Pregnant and Postnatal Mothers

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Introduction

- Pregnancy requires continuous maternal health monitoring.
- Postnatal period involves recovery and hidden health risks.
- Delayed diagnosis increases maternal complications.
- Wireless IoT wearables enable real-time physiological monitoring.
- Deep learning supports early risk prediction.
- Wireless alert mechanisms ensure timely medical response.

This project unifies pregnancy and postnatal care in one system.

Problem Statement

UNDETECTED RISKS

Health risks often remain unseen between hospital visits.

INADEQUATE MONITORING

Lack of continuous wireless systems delays diagnosis.

FRAGMENTED CARE

Focus on pregnancy OR postnatal, not both periods.

RELIANCE ON SELF-REPORTING

Depends on subjective input and limited hospital access.

DELAYED EMERGENCE

Slow response time during critical situations.

LIMITED UNIFIED SOLUTIONS

Smart, wireless, all-in-one systems lacking in rural areas.

Figure 1. Key challenges in pregnancy and postnatal care motivating a smart wireless IoT system.

| S/N | Paper & Author(s)                                                                   | Key Contribution                                                                              | Observation                                                        |
|-----|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| 1   | Malde et al. (2025)<br><i>Maternal Health Risk Prediction Using Vital Signs</i>     | Predicts maternal risk (Low/Medium/High) using limited vital-sign data with machine learning. | Uses offline data only; no real-time monitoring or alert system.   |
| 2   | Tzimourta et al. (2025)<br><i>AI-Based Maternal Risk Detection</i>                  | Applies ML models for pregnancy risk classification with good accuracy.                       | Prediction-only approach; lacks IoT implementation and deployment. |
| 3   | Liu et al. (2024)<br><i>Wearable Sensors and AI in Pregnancy Monitoring</i>         | Reviews wearable sensors and AI pipelines for pregnancy monitoring.                           | No complete system or real-world implementation proposed.          |
| 4   | Rattanasak et al. (2025)<br><i>Wearable IoT System for Fetal Movement Detection</i> | Develops an ESP32-based wearable with real-time sensor data and ML.                           | Focuses on fetal movement only; maternal risk not addressed.       |
| 5   | Milan et al. (2024)<br><i>Wearable Signals and Pregnancy Physiology</i>             | Shows correlation between wearable sensor signals and pregnancy changes.                      | No risk prediction, alerts, or integrated healthcare system.       |

Research Gap

Offline Data

Focus mainly on offline analysis or periodic clinical assessments.

No Actionable AI

AI models classify risk but do not trigger real-time actions.

Passive Wearables

Devices collect data but lack intelligent decision-making.

No Alerts

No automatic alert or quick response during emergencies.

Postnatal Neglect

Post-delivery maternal care is often ignored.

Integration Gap

No unified IoT hardware, deep learning, wireless alerts.

Figure 2. Research Gap in Maternal Healthcare Monitoring

Objectives

Postnatal Maternal Care

Recovery risks, Vital trend

Continuous Vital Sign Monitoring

ECG, HR, HRV, SpO<sub>2</sub>, temperature, motion

Multi-Level Alert Mechanism

Buzzer, LED, GSM SMS, GPS location sharing

Low-Cost, Portable & Offline Usability

Suitable for rural and resource-constrained areas

Deep Learning–Based Risk Prediction

Normal, Warning, and Critical condition classification

Wireless Data Transmission

Wi-Fi, Bluetooth, GSM for remote monitoring

Figure 3. Project objectives for Materna.

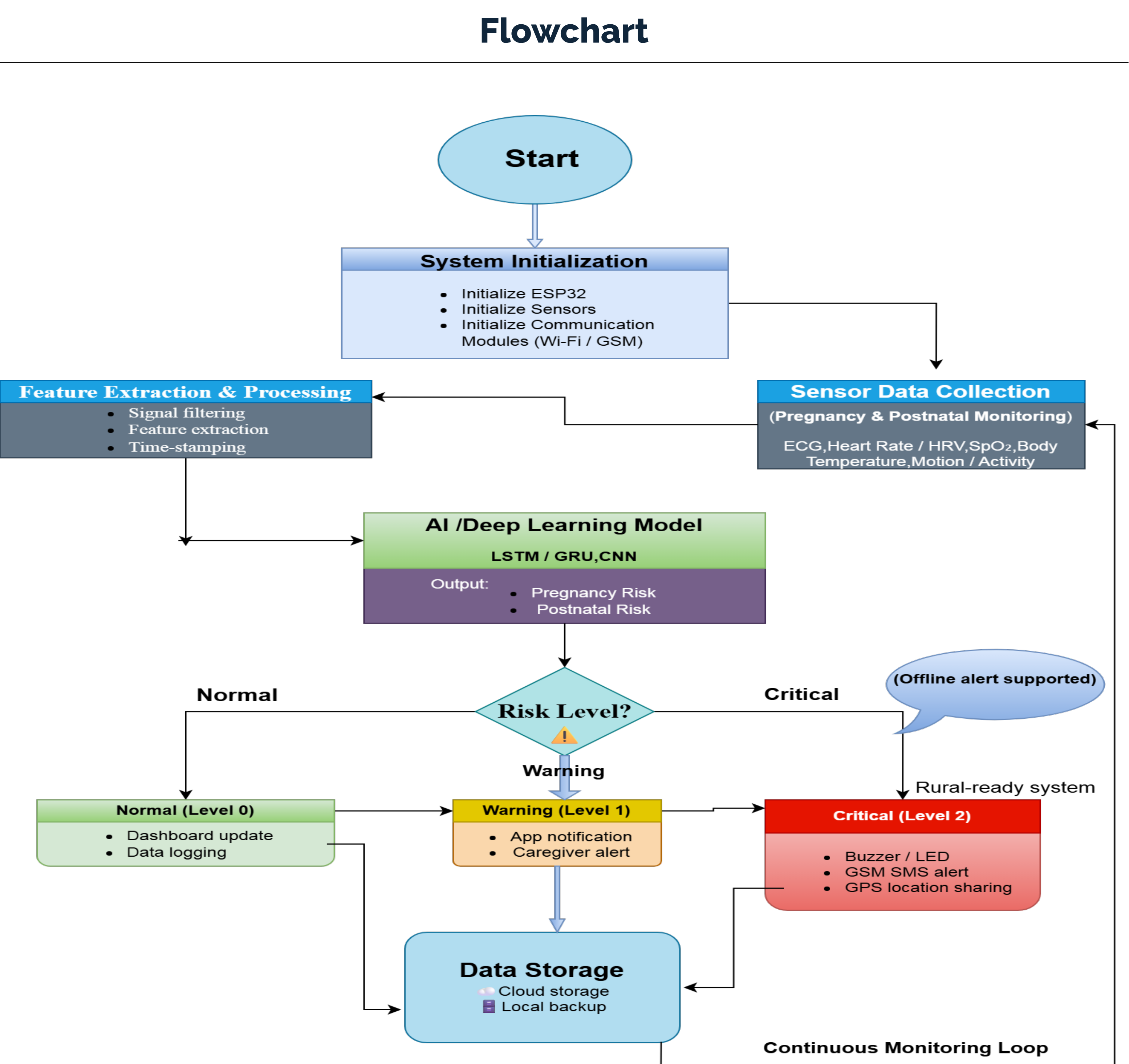


Figure 4. Workflow of the Proposed Maternal Health Monitoring System

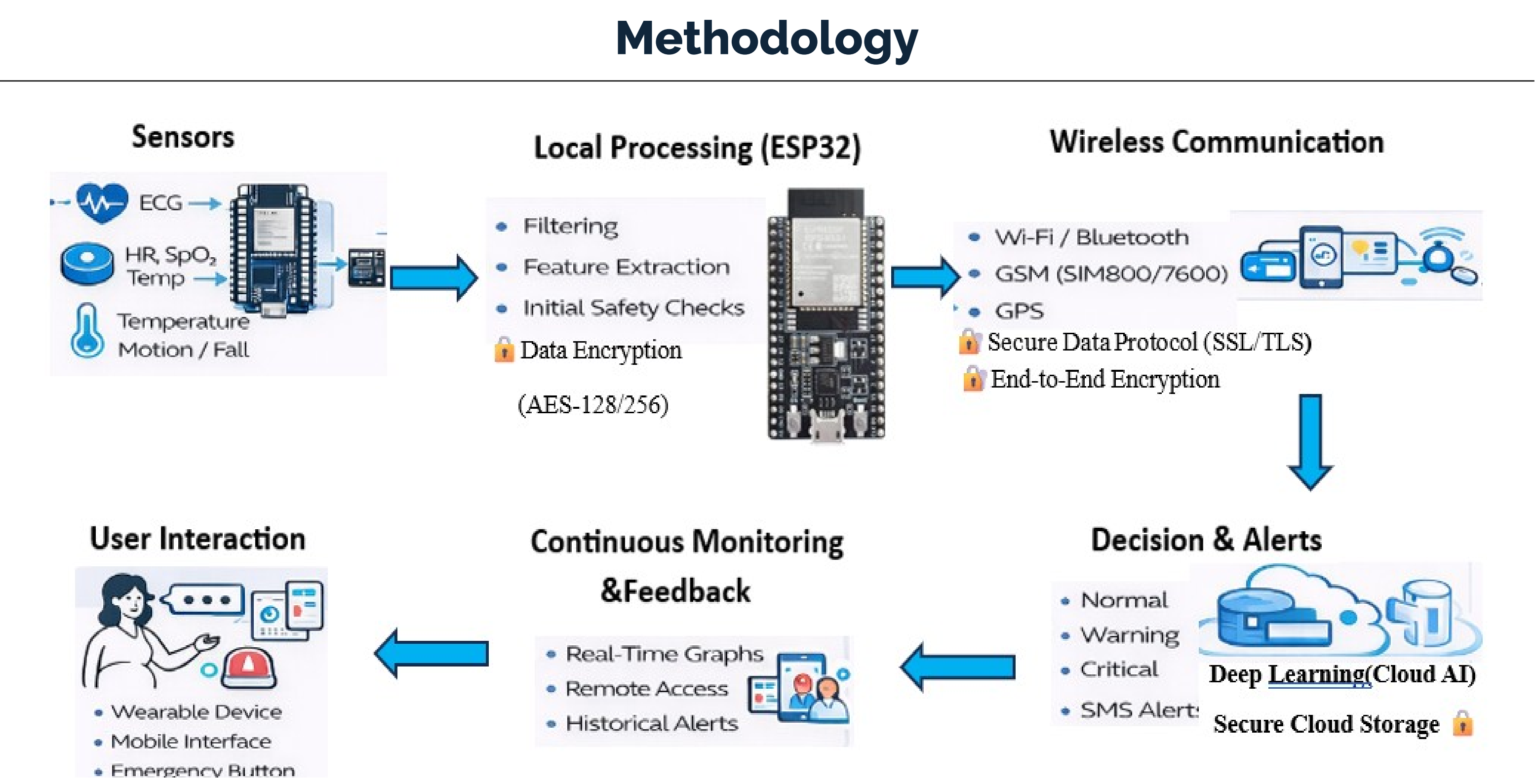


Figure 5. Proposed system architecture for maternal health monitoring

Expected Outcome

01

Real-Time Maternal Health Monitoring

Continuous real-time monitoring of maternal vital signs including ECG, heart rate, HRV, SpO<sub>2</sub>, body temperature, and physical activity during pregnancy and postnatal periods.

02

Early Risk Prediction Using AI

Deep learning models (LSTM/GRU/CNN) for early detection of maternal health risks.

03

Multi-Level Alert and Emergency Response

• Normal: Dashboard update  
• Warning: App alert  
• Critical: Buzzer/LED, SMS, GPS

04

Pregnancy and Postnatal Care Support

Continuous monitoring of pregnancy and postnatal health risks.

05

Secure Data Logging and Visualization

Maternal health data stored securely in cloud and local storage for analysis and display.

06

Rural-Ready and Scalable Healthcare System

Low-cost, offline-capable IoT solution for rural and resource-limited areas.

Figure 6. Expected Outcomes of the Proposed System

Conclusion

Materna Highlights

- Real-time IoT-based maternal health monitoring.
- AI-driven pregnancy and postnatal risk prediction.
- Multi-level alerts with emergency response support.
- Continuous data logging and remote monitoring.
- Scalable, low-cost system for rural healthcare.

References

IRE 300 : Capstone Project

University of Frontier Technology,Bangladesh(UFTB)

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